

Data from each sensor is filtered to remove noise and is often fused with other data sources to augment the original image. The vehicle subsequently uses this data to make navigation decisions, determined by its control system.

The majority of self-driving vehicle control systems implement a deliberative architecture, meaning that these are capable of making intelligent decisions by maintaining an internal map of their world and by using that map to find an optimal path to their destination that avoids obstacles (for example, road structures, pedestrians and other vehicles) from a set of possible paths. Once the vehicle determines the best path to take, the decision is dissected into commands, which are fed to the vehicle's actuators.

These actuators control the vehicle's steering, braking and throttle. This process of localisation, mapping, obstacle avoidance and path planning is repeated multiple times each second on powerful onboard processors until the vehicle reaches its destination.

Mapping and localisation

Prior to making any navigation decisions, the vehicle must first build a map of its environment and precisely localise itself within that map. The most frequently-used sensors for map building are laser range-finders and cameras. Laser range-finders scan the environment using swaths of laser beams and calculate the distance to nearby objects by measuring the time it takes for each laser beam to travel to the object and back.

Where video from camera is ideal for extracting scene colour, with laser range-finders depth information is readily available to the vehicle for building a 3D map. Because

laser beams diverge as these travel through Space, it is difficult to obtain accurate distance readings from more than 100 metres away using the most state-of-the-art laser range-finders, which limit the amount of reliable data that can be captured on the map. The vehicle filters and discretises data collected from each sensor and often aggregates the information to create a comprehensive



Fig. 3: Autonomous vehicles

map, which can then be used for path planning.

For the vehicle to know where it is in relation to other objects on the map, it must use its GPS, inertial navigation unit and sensors to precisely localise itself. GPS estimates can be off by many metres due to signal delays caused by changes in the atmosphere and reflections off nearby buildings and surrounding terrain, and inertial navigation units accumulate position errors over time. Therefore localisation algorithms often incorporate maps or sensor data previously collected from the same location to reduce uncertainty. As the vehicle moves, new positional information and sensor data are used to update the vehicle's internal map.

Obstacle avoidance

A vehicle's internal map includes the current and predicted location of

all static (buildings, traffic lights and stop signs) and moving (other vehicles and pedestrians) obstacles in its vicinity. Obstacles are categorised depending on how well these match with a library of pre-determined shape and motion descriptors.

The vehicle uses a probabilistic model to track the predicted future path of moving objects based on its shape and prior trajectory. For example, if a two-wheeled object is travelling at 60 kilometres per hour versus 15 kilometres per hour, it is most likely a motorcycle and not a bicycle, and will get categorised as such by the vehicle. This process allows the vehicle to make more intelligent decisions when approaching crosswalks or busy intersections. The previous, current and predicted future locations of all

obstacles in the vehicle's vicinity are incorporated into its internal map, which the vehicle then uses to plan its path.

Path planning

The goal of path planning is to use the information captured in the vehicle's map to safely direct the vehicle to its destination while avoiding obstacles and following the rules of the road. Although vehicle manufacturers' planning algorithms will be different based on their navigation objectives and sensors used, the following describes a general path-planning algorithm that has been used on military ground vehicles.

The algorithm determines a rough long-range plan for the vehicle to follow while continuously refining a short-range plan (for example, change lanes, drive forward ten metres and turn right).